

OPTIMAL INVISCID DAMPING AND VORTICITY DEPLETION FOR NON-MONOTONIC SHEAR FLOWS

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APPENDIX A. EXPLICIT RAYLEIGH EQUATIONS

In this appendix, we display explicit formulas for $\mathcal{R}_{a,b,\epsilon}^\iota$, the remainder terms obtained by applying the Rayleigh operator to higher-order derivatives of the spectral density functions after subtracting the corresponding singular terms $\mathcal{S}_{a,b,\epsilon}^\iota$. The table below lists the lemmas whose proofs use these formulas.

$\mathcal{R}_{a,b,\epsilon}^\iota$	Non-Degenerate	Degenerate
$\mathcal{R}_{1,0,\epsilon}^\iota$	Lemma 5.7	Lemma 7.3
$\mathcal{R}_{2,0,\epsilon}^\iota$	Lemma 5.9	Lemma 7.5
$\mathcal{R}_{1,1,\epsilon}^\iota$	Lemma 5.10	Lemma 7.7
$\mathcal{R}_{3,0,\epsilon}^\iota$	Lemma 5.12	Lemma 7.11
$\mathcal{R}_{2,1,\epsilon}^\iota$	Lemma 5.14	Lemma 7.12
$\mathcal{R}_{1,2,\epsilon}^\iota$	Lemma 5.15	Lemma 7.13

A.1. Non-degenerate case. In the nondegenerate case, the spectral density function $\psi_{k,\epsilon}^\iota$ is defined by (2.6). Its higher-order spectral derivatives satisfy

$$\left(\partial_y^2 - k^2 - \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) \left(\partial_\lambda^a D_\lambda^b \psi_{k,\epsilon}^\iota - \mathcal{S}_{a,b,\epsilon}^\iota \right) = \mathcal{R}_{a,b,\epsilon}^\iota(y, \lambda).$$

As in the proofs, we group the terms in $\mathcal{R}_{a,b,\epsilon}^\iota$ according to their singularity structure.

Throughout this subsection, we use the following shorthand notations:

$$\psi = \psi_{k,\epsilon}^\iota, \quad b = b(y), \quad \chi = \chi_{*,\delta_0/10}.$$

(1) $\mathcal{R}_{1,0,\epsilon}^\iota(y, \lambda)$ appearing in (5.27) and (5.28) has the explicit representation

$$\mathcal{R}_{1,0,\epsilon}^\iota(y, \lambda) = c_1^{(1,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,0)}(y) + \mathfrak{c}^{(1,0)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} + c_3^{(1,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,0)}(y), \quad (\text{A.1})$$

with

$$\begin{aligned} c_1^{(1,0)}(y) &= 0, \\ c_2^{(1,0)}(y, \lambda) &= \left[2b' \partial_y \left(\frac{1-\chi}{(b')^2} \right) + \frac{(1-\chi)b''}{(b')^2} \right] (b''\psi - \omega_{0k}) + \frac{2(1-\chi)}{b'} (b'''\psi - \omega'_{0k}), \\ \mathfrak{c}^{(1,0)}(y) &= \frac{2(1-\chi)b''}{b'}, \\ c_3^{(1,0)}(y, \lambda) &= \partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) \psi + 2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) \partial_y \psi - \partial_y^2 \left(\frac{1-\chi}{(b')^2} \right) \omega_{0k} - 2\partial_y \left(\frac{1-\chi}{(b')^2} \right) \omega'_{0k} \end{aligned}$$

$$+ \frac{1-\chi}{(b')^2}(k^2\omega_{0k} - \omega_{0k}''),$$

$$c_4^{(1,0)}(y, \lambda) = \chi \frac{b''\psi - \omega_{0k}}{(b(y) - \lambda + i\epsilon)^2}$$

(2) $\mathcal{R}_{2,0,\epsilon}'(y, \lambda)$ appearing in (5.35) and (5.36) has the explicit representation

$$\begin{aligned} \mathcal{R}_{2,0,\epsilon}'(y, \lambda) = & c_1^{(2,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(2,0)}(y) + \mathfrak{c}_1^{(2,0)}(y)\partial_y D_\lambda \psi + \mathfrak{c}_2^{(2,0)}(y)\partial_y \psi}{b(y) - \lambda + i\epsilon} \\ & + c_3^{(2,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(2,0)}(y), \end{aligned} \quad (\text{A.2})$$

with

$$\begin{aligned} c_1^{(2,0)}(y, \lambda) = & (1 - \chi) \frac{2[b'b''' - 2(b'')^2]}{(b')^4} (b''\psi - \omega_{0k}), \\ c_2^{(2,0)}(y, \lambda) = & \left\{ -\partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) - 2b'\partial_y \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] - (1-\chi)b'' \left[-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right] \right\} \psi \\ & + \left[4b'\partial_y \left(\frac{(1-\chi)b''}{(b')^3} \right) + \frac{2(1-\chi)(b'')^2}{(b')^3} \right] D_\lambda \psi + (\partial_y^2 - k^2) \left(\frac{(1-\chi)\omega_{0k}}{(b')^2} \right) \\ & - 2b'\partial_y \left[(1-\chi) \left(\frac{2\omega_{0k}'}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right) \right] - (1-\chi)b'' \left[\frac{2\omega_{0k}'}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right], \\ \mathfrak{c}_1^{(2,0)}(y) = & \frac{4(1-\chi)b''}{(b')^2}, \\ \mathfrak{c}_2^{(2,0)}(y) = & -2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) + (1-\chi) \left[\frac{4b'''}{(b')^2} - \frac{6(b'')^2}{(b')^3} \right], \\ c_3^{(2,0)}(y, \lambda) = & -(\partial_y^2 - k^2) \left[(1-\chi) \left(\frac{2\omega_{0k}'}{(b')^3} - \frac{3b''\omega_{0k}}{(b')^4} \right) \right] \\ & - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] + \frac{4k^2(1-\chi)(b'')^2}{(b')^4} \right\} \psi \\ & - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{2b'''}{(b')^3} + \frac{3(b'')^2}{(b')^4} \right) \right] + \frac{2(1-\chi)b''}{(b')^3} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} \partial_y \psi \\ & + \left\{ \partial_y^2 \left(\frac{2(1-\chi)b''}{(b')^3} \right) + \frac{2(1-\chi)b''}{(b')^3} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} D_\lambda \psi \\ & + \left[4\partial_y \left(\frac{(1-\chi)b''}{(b')^3} \right) + \frac{4(1-\chi)(b'')^2}{(b')^4} \right] \partial_y D_\lambda \psi, \\ c_4^{(2,0)}(y, \lambda) = & \chi \left[\frac{2(b''\psi - \omega_{0k})}{(b(y) - \lambda + i\epsilon)^3} + \frac{2b''\partial_\lambda \psi}{(b(y) - \lambda + i\epsilon)^2} \right] - \chi' \frac{2(b''\psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^2}. \end{aligned}$$

(3) $\mathcal{R}_{1,1,\epsilon}'(y, \lambda)$ appearing in (5.41) and (5.42) has the explicit representation

$$\begin{aligned} \mathcal{R}_{1,1,\epsilon}'(y, \lambda) = & c_1^{(1,1)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,1)}(y) + \mathfrak{c}_1^{(1,1)}(y)\partial_y D_\lambda \psi + \mathfrak{c}_2^{(1,1)}(y)\partial_y \psi}{b(y) - \lambda + i\epsilon} \\ & + c_3^{(1,1)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,1)}(y) \end{aligned} \quad (\text{A.3})$$

with

$$\begin{aligned} c_1^{(1,1)}(y, \lambda) = & (1 - \chi) \frac{b'b''' - 2(b'')^2}{(b')^3} (b''\psi - \omega_{0k}), \\ c_2^{(1,1)}(y, \lambda) = & \left\{ \partial_y^2 \left(\frac{(1-\chi)b''}{b'} \right) - 2b'\partial_y \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] - (1-\chi)b'' \left[\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right] \right\} \psi \end{aligned}$$

$$\begin{aligned}
& + \left[\frac{(1-\chi)b'''}{b'} - 2b'\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) - \frac{(1-\chi)(b'')^2}{(b')^2} \right] D_\lambda \psi - (\partial_y^2 - k^2) \left(\frac{(1-\chi)\omega_{0k}}{b'} \right) \\
& + 2b'\partial_y \left[(1-\chi) \left(\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right) \right] + (1-\chi)b'' \left[\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right] + b'c_2^{(2,0)}, \\
c_1^{(1,1)}(y) &= \frac{2(1-\chi)b''}{b'}, \\
c_2^{(1,1)}(y) &= (1-\chi) \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right], \\
c_3^{(1,1)}(y, \lambda) &= b'c_3^{(2,0)} + (\partial_y^2 - k^2) \left[(1-\chi) \left(\frac{\omega'_{0k}}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} \right) \right] \\
& - \left\{ \partial_y^2 \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] - \frac{2k^2(1-\chi)(b'')^2}{(b')^3} \right\} \psi \\
& + \left\{ -2\partial_y \left[(1-\chi) \left(\frac{b'''}{(b')^2} - \frac{(b'')^2}{(b')^3} \right) \right] + \frac{(1-\chi)b''}{(b')^2} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} \partial_y \psi \\
& - \left\{ \partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) + \frac{(1-\chi)b''}{(b')^2} \left[\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right] \right\} D_\lambda \psi \\
& - \left[2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) + \frac{2(1-\chi)(b'')^2}{(b')^3} \right] \partial_y D_\lambda \psi, \\
c_4^{(1,1)}(y, \lambda) &= \chi \left[\frac{b''D_\lambda \psi + b'''\psi - \omega'_{0k}}{(b(y) - \lambda + i\epsilon)^2} + \frac{b'''\partial_\lambda \psi}{b(y) - \lambda + i\epsilon} \right] + 2b''\partial_y r_{2,0,\epsilon}^L + b'''\partial_y r_{2,0,\epsilon}^L.
\end{aligned}$$

(4) $\mathcal{R}_{3,0,\epsilon}^L(y, \lambda)$ appearing in (5.49) and (5.50) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{3,0,\epsilon}^L(y, \lambda) &= c_1^{(3,0)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(3,0)}(y) + c_1^{(3,0)}(y)\partial_y D_\lambda^2 \psi + c_2^{(3,0)}(y)\partial_y D_\lambda \psi + c_3^{(3,0)}(y)\partial_y \psi}{b(y) - \lambda + i\epsilon} \\
& + c_3^{(3,0)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(3,0)}(y)
\end{aligned} \tag{A.4}$$

with

$$\begin{aligned}
c_1^{(3,0)}(y, \lambda) &= (1-\chi) \left[\frac{3b''b^{(4)} + 6(b''')^2}{(b')^4} - \frac{42(b'')^2b'''}{(b')^5} + \frac{42(b'')^4}{(b')^6} \right] \psi \\
& + (1-\chi) \left[-\frac{3b^{(4)}}{(b')^4} + \frac{30b''b'''}{(b')^5} - \frac{42(b'')^3}{(b')^6} \right] \omega_{0k} + (1-\chi) \left[-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right] \omega'_{0k} \\
& + (1-\chi) \left[\frac{6b''b'''}{(b')^4} - \frac{12(b'')^3}{(b')^5} \right] D_\lambda \psi \\
c_2^{(3,0)}(y, \lambda) &= - \left\{ 2k^2\partial_y \left[-\frac{(1-\chi)b''}{(b')^3} \right] + \partial_y^2 \left[(1-\chi) \left(\frac{2b'''}{(b')^3} - \frac{3(b'')^2}{(b')^4} \right) \right] \right. \\
& - k^2 \left(\frac{2b''}{b'} \right) \left(\frac{3(1-\chi)b''}{(b')^3} \right) \\
& + 2b'\partial_y \left[(1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right] \\
& + b'' \left((1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right) \left. \right\} \psi \\
& - \left\{ \partial_y^2 \left[\frac{3(1-\chi)b''}{(b')^3} \right] + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left(\frac{3(1-\chi)b''}{(b')^3} \right) \right. \\
& + 2b'\partial_y \left[(1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \left. \right\}
\end{aligned}$$

$$\begin{aligned}
& + b'' \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \Big\} D_\lambda \psi \\
& - \left\{ 2b' \partial_y \left[-\frac{3(1-\chi)b''}{(b')^4} \right] + b'' \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} D_\lambda^2 \psi \\
& - \left[\left(-\frac{2(1-\chi)}{(b')^3} \right) + 2b' \left(\frac{3(1-\chi)}{(b')^4} \right) \right] \omega_{0k}''' \\
& - \left\{ 2\partial_y \left[-\frac{2(1-\chi)}{(b')^3} \right] + \left(\frac{3(1-\chi)b''}{(b')^4} \right) \right. \\
& \quad \left. + 2b' \left[\left(-\frac{12(1-\chi)b''}{(b')^5} \right) + \partial_y \left[\frac{3(1-\chi)}{(b')^4} \right] \right] + b'' \left(\frac{3(1-\chi)}{(b')^4} \right) \right\} \omega_{0k}'' \\
& - \left\{ \partial_y^2 \left[-\frac{2(1-\chi)}{(b')^3} \right] - k^2 \left(-\frac{2(1-\chi)}{(b')^3} \right) + 2\partial_y \left[\frac{3(1-\chi)b''}{(b')^4} \right] \right. \\
& \quad \left. + 2b' \left[\partial_y \left[-\frac{12(1-\chi)b''}{(b')^5} \right] + \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right] \right. \\
& \quad \left. + b'' \left(-\frac{12(1-\chi)b''}{(b')^5} \right) \right\} \omega_{0k}' \\
& - \left\{ \partial_y^2 \left[\frac{3(1-\chi)b''}{(b')^4} \right] - k^2 \left(\frac{3(1-\chi)b''}{(b')^4} \right) \right. \\
& \quad \left. + 2b' \partial_y \left[(1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \right. \\
& \quad \left. + b'' \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \right\} \omega_{0k}, \\
\mathfrak{c}_1^{(3,0)}(y) &= -2b' \left(-\frac{3(1-\chi)b''}{(b')^4} \right), \\
\mathfrak{c}_2^{(3,0)}(y) &= -2\partial_y \left[\frac{3(1-\chi)b''}{(b')^3} \right] - \left(\frac{2b''}{b'} \right) \left(\frac{3(1-\chi)b''}{(b')^3} \right) - 2b' \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right), \\
\mathfrak{c}_3^{(3,0)}(y) &= -\partial_y^2 \left[-\frac{(1-\chi)b''}{(b')^3} \right] - 2\partial_y \left[(1-\chi) \left(\frac{2b'''}{(b')^3} - \frac{3(b'')^2}{(b')^4} \right) \right] \\
& \quad + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left(\frac{3(1-\chi)b''}{(b')^3} \right) \\
& \quad - 2b' \left((1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right), \\
c_3^{(3,0)}(y) &= - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right] \right. \\
& \quad - k^2 \left(\frac{2b''}{b'} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \\
& \quad \left. - k^2 \left(\frac{2b'''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} \psi \\
& - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{3b^{(4)}}{(b')^4} + \frac{16b''b'''}{(b')^5} - \frac{15(b'')^3}{(b')^6} - \frac{k^2b''}{(b')^4} \right) \right] \right. \\
& \quad \left. - \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \right\}
\end{aligned}$$

$$\begin{aligned}
& - \left(\frac{b^{(4)}}{b'} - \frac{6b''b'''}{(b')^2} + \frac{6(b'')^3}{(b')^3} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \Big\} \partial_y \psi \\
& - \left\{ \partial_y^2 \left[(1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \right. \\
& \quad + \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \\
& \quad + \left(\left(\frac{b^{(4)}}{b'} - \frac{6b''b'''}{(b')^2} + \frac{6(b'')^3}{(b')^3} \right) - 2k^2 \left(\frac{2b''}{b'} \right) \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \Big\} D_\lambda \psi \\
& - \left\{ 2\partial_y \left[(1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right] \right. \\
& \quad + \left(\frac{2b''}{b'} \right) \left((1-\chi) \left(-\frac{6b'''}{(b')^4} + \frac{12(b'')^2}{(b')^5} \right) \right) \Big\} \partial_y D_\lambda \psi \\
& - \left\{ \partial_y^2 \left[-\frac{3(1-\chi)b''}{(b')^4} \right] + \left(\frac{2b'''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} D_\lambda^2 \psi \\
& - \left\{ 2\partial_y \left[-\frac{3(1-\chi)b''}{(b')^4} \right] + 2 \left(\frac{2b''}{b'} \right) \left(-\frac{3(1-\chi)b''}{(b')^4} \right) \right\} \partial_y D_\lambda^2 \psi \\
& - \left(\frac{3(1-\chi)}{(b')^4} \right) \omega_{0k}^{(4)} \\
& - \left(\left(-\frac{12(1-\chi)b''}{(b')^5} \right) + 2\partial_y \left[\frac{3(1-\chi)}{(b')^4} \right] \right) \omega_{0k}''' \\
& - \left\{ 2\partial_y \left[-\frac{12(1-\chi)b''}{(b')^5} \right] \right. \\
& \quad + \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \\
& \quad + \partial_y^2 \left[\frac{3(1-\chi)}{(b')^4} \right] - k^2 \left(\frac{3(1-\chi)}{(b')^4} \right) \Big\} \omega_{0k}'' \\
& - \left\{ \partial_y^2 \left[-\frac{12(1-\chi)b''}{(b')^5} \right] - k^2 \left(-\frac{12(1-\chi)b''}{(b')^5} \right) \right. \\
& \quad + 2\partial_y \left[(1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \Big\} \omega_{0k}' \\
& - \left\{ \partial_y^2 \left[(1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right] \right. \\
& \quad - k^2 \left((1-\chi) \left(\frac{k^2}{(b')^4} - \frac{4b'''}{(b')^5} + \frac{15(b'')^2}{(b')^6} \right) \right) \Big\} \omega_{0k}, \\
& c_4^{(3,0)}(y) = \chi \left[\frac{6(b''\psi - \omega_{0k})}{(b(y) - \lambda + i\epsilon)^4} + \frac{6b''\partial_\lambda \psi}{(b(y) - \lambda + i\epsilon)^3} + \frac{3b''\partial_\lambda^2 \psi}{(b(y) - \lambda + i\epsilon)^2} \right] \\
& + \chi' \left\{ -\frac{4(b''\psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^3} + \frac{2}{(b(y) - \lambda + i\epsilon)^2} \left[\left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \psi \right. \right. \\
& \quad \left. \left. + \frac{2b''}{(b')^2} \partial_y \psi + \frac{\omega_{0k}'}{(b')^2} - \frac{b''\omega_{0k}}{(b')^3} - \frac{3b''}{(b')^2} D_\lambda \psi \right] \right\}
\end{aligned}$$

$$+ \chi'' \frac{b''\psi - \omega_{0k}}{(b')^2(b(y) - \lambda + i\epsilon)^2}.$$

(5) $\mathcal{R}_{2,1,\epsilon}^{\iota}(y, \lambda)$ appearing in (5.55) and (5.56) has the explicit representation

$$\begin{aligned} \mathcal{R}_{2,1,\epsilon}^{\iota}(y, \lambda) = & c_1^{(2,1)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(2,1)}(y) + c_1^{(2,1)}(y) \partial_y D_{\lambda}^2 \psi + c_2^{(2,1)}(y) \partial_y D_{\lambda} \psi + c_3^{(2,1)}(y) \partial_y \psi}{b(y) - \lambda + i\epsilon} \\ & + c_3^{(2,1)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(2,1)}(y), \end{aligned} \quad (\text{A.5})$$

with

$$\begin{aligned} c_1^{(2,1)}(y, \lambda) = & b' c_1^{(3,0)} - \left[(1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \left(b''' - \frac{2(b'')^2}{b'} \right) + \frac{(1 - \chi)b''}{(b')^3} \left(b^{(4)} - \frac{6b''b'''}{b'} \right. \right. \\ & \left. \left. + \frac{6(b'')^3}{(b')^2} \right) \right] \psi - \frac{(1 - \chi)b''}{(b')^3} \left(2b''' - \frac{4(b'')^2}{b'} \right) D_{\lambda} \psi + \left(b'' \frac{3(1 - \chi)b''}{(b')^4} \right. \\ & \left. + (1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} - \frac{4b''}{b'} \frac{(1 - \chi)b''}{(b')^3} \right) \omega'_{0k} \\ & + \left[(1 - \chi) \frac{k^2(b')^2 b'' + (b')^2 b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right. \\ & \left. + b'' \left[-(1 - \chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] - \frac{2b''}{b'} (1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right. \\ & \left. - \left(\frac{2b'''}{b'} - \frac{6(b'')^2}{(b')^2} \right) \frac{(1 - \chi)b''}{(b')^3} \right] \omega_{0k}, \\ c_2^{(2,1)}(y) = & b' c_2^{(3,0)} + \left(\frac{(1 - \chi)b'''}{(b')^2} - 2b' \partial_y \left[\frac{(1 - \chi)b''}{(b')^3} \right] - b'' \frac{(1 - \chi)b''}{(b')^3} \right) D_{\lambda}^2 \psi \\ & + \left[\frac{(1 - \chi)b''b'''}{(b')^3} + 2 \frac{(1 - \chi)b''}{(b')^2} \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) + 2\partial_y^2 \left[\frac{(1 - \chi)b''}{(b')^2} \right] \right. \\ & \left. - 2b' \partial_y \left[(1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] - b'' (1 - \chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] D_{\lambda} \psi \\ & + \left[\frac{k^2(1 - \chi)b'''}{(b')^2} - 2k^2 \partial_y \left[\frac{(1 - \chi)b''}{(b')^2} \right] - \partial_y^2 \left[(1 - \chi) \left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \right] \right. \\ & \left. - \frac{4k^2 b''}{b'} \frac{(1 - \chi)b''}{(b')^2} - 2b' \partial_y \left[(1 - \chi) \frac{k^2(b')^2 b'' + (b')^2 b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \right. \\ & \left. - b'' (1 - \chi) \frac{k^2(b')^2 b'' + (b')^2 b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \psi + \left(-\frac{1 - \chi}{(b')^2} \right. \\ & \left. + 2b' \frac{1 - \chi}{(b')^3} \right) \omega'_{0k} + \left(-2\partial_y \left[\frac{1 - \chi}{(b')^2} \right] + \frac{(1 - \chi)b''}{(b')^3} + 2b' \partial_y \left[\frac{1 - \chi}{(b')^3} \right] \right. \\ & \left. - 2b' \frac{3(1 - \chi)b''}{(b')^4} + b'' \frac{1 - \chi}{(b')^3} \right) \omega_{0k} + \left(k^2 \frac{1 - \chi}{(b')^2} - \partial_y^2 \left[\frac{1 - \chi}{(b')^2} \right] \right. \\ & \left. + 2\partial_y \left[\frac{(1 - \chi)b''}{(b')^3} \right] - 2b' \partial_y \left[\frac{3(1 - \chi)b''}{(b')^4} \right] \right. \\ & \left. - 2b' \left[-(1 - \chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] - b'' \frac{3(1 - \chi)b''}{(b')^4} \right) \omega'_{0k} + \left(\right. \\ & \left. - k^2 \frac{(1 - \chi)b''}{(b')^3} + \partial_y^2 \left[\frac{(1 - \chi)b''}{(b')^3} \right] \right. \\ & \left. - 2b' \partial_y \left[-(1 - \chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \right. \\ & \left. - b'' \left[-(1 - \chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \right) \omega_{0k}, \end{aligned}$$

$$\begin{aligned}
\mathfrak{c}_1^{(2,1)}(y) &= b' \mathfrak{c}_1^{(3,0)} - 2b' \frac{(1-\chi)b''}{(b')^3}, \\
\mathfrak{c}_2^{(2,1)}(y) &= b' \mathfrak{c}_2^{(3,0)} - \frac{2(1-\chi)b'''}{(b')^2} + \frac{4b''}{b'} \frac{(1-\chi)b''}{(b')^2} + 4\partial_y \left[\frac{(1-\chi)b''}{(b')^2} \right] \\
&\quad - 2b'(1-\chi) \frac{2b'b''' - 3(b'')^2}{(b')^4}, \\
\mathfrak{c}_3^{(2,1)}(y) &= b' \mathfrak{c}_3^{(3,0)} - \frac{(1-\chi)b''b'''}{(b')^3} - \partial_y^2 \left[\frac{(1-\chi)b''}{(b')^2} \right] - 2\partial_y \left[(1-\chi) \left(-\frac{b'''}{(b')^2} + \frac{(b'')^2}{(b')^3} \right) \right] \\
&\quad - 2 \frac{(1-\chi)b''}{(b')^2} \left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) \\
&\quad - 2b'(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5}, \\
c_3^{(2,1)}(y) &= b' \mathfrak{c}_3^{(3,0)} - \frac{3(1-\chi)b''}{(b')^4} (\omega_{0k}''' - k^2 \omega_{0k}') - 2\partial_y \left[\frac{3(1-\chi)b''}{(b')^4} \right] \omega_{0k}'' \\
&\quad - \partial_y^2 \left[\frac{3(1-\chi)b''}{(b')^4} \right] \omega_{0k}' \\
&\quad - 2\partial_y \left[(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \partial_y \psi \\
&\quad - \partial_y^2 \left[(1-\chi) \frac{k^2(b')^2b'' + (b')^2b^{(4)} - 4b'b''b''' + 3(b'')^3}{(b')^5} \right] \psi \\
&\quad - \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] (\omega_{0k}'' - k^2 \omega_{0k}') \\
&\quad - 2\partial_y \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \omega_{0k}' \\
&\quad - \partial_y^2 \left[-(1-\chi) \frac{k^2(b')^2 - b'b''' + 3(b'')^2}{(b')^5} \right] \omega_{0k} \\
&\quad - (1-\chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \left[\left(\frac{b'''}{b'} - \frac{2(b'')^2}{(b')^2} \right) (D_\lambda \psi - \partial_y \psi) \right. \\
&\quad \left. + \frac{2b''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \right] - 2\partial_y \left[(1-\chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] \partial_y D_\lambda \psi \\
&\quad - \partial_y^2 \left[(1-\chi) \frac{2b'b''' - 3(b'')^2}{(b')^4} \right] D_\lambda \psi \\
&\quad - \frac{4b''}{b'} \frac{(1-\chi)b''}{(b')^3} \left[\frac{\partial_y D_\lambda^2 \psi - k^2 D_\lambda \psi}{b'} + \left(-\frac{b'''}{b'} + \frac{2(b'')^2}{(b')^2} \right) (D_\lambda \psi - \partial_y \psi) \right] \\
&\quad \quad - \frac{2b''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \\
&\quad - \frac{(1-\chi)b''}{(b')^3} \left[\left(-\frac{4(b'')^2}{(b')^2} + \frac{2b'''}{b'} \right) (D_\lambda^2 \psi - \partial_y D_\lambda \psi) + \frac{2b'''}{b'} (\partial_y D_\lambda \psi - k^2 \psi) \right] \\
&\quad + \frac{2(b'')^2}{(b')^2} \frac{(1-\chi)b''}{(b')^3} \left[D_\lambda^2 \psi - 2\partial_y D_\lambda \psi + k^2 \psi + \frac{b''}{b'} (D_\lambda \psi - \partial_y \psi) \right] \\
&\quad - \frac{(1-\chi)b''}{(b')^3} \left(-\frac{2b''b'''}{(b')^2} + \frac{b^{(4)}}{b'} \right) (D_\lambda \psi - \partial_y \psi) - 2\partial_y \left[\frac{(1-\chi)b''}{(b')^3} \right] \partial_y D_\lambda^2 \psi \\
&\quad - \partial_y^2 \left[\frac{(1-\chi)b''}{(b')^3} \right] D_\lambda^2 \psi + \frac{1-\chi}{(b')^3} (\omega_{0k}^{(4)} - k^2 \omega_{0k}'') + 2\partial_y \left[\frac{1-\chi}{(b')^3} \right] \omega_{0k}''' \\
&\quad + \partial_y^2 \left[\frac{1-\chi}{(b')^3} \right] \omega_{0k}'',
\end{aligned}$$

$$\begin{aligned}
c_4^{(2,1)}(y) &= 2b''\partial_y r_{3,0,\epsilon}^t + b'''r_{3,0,\epsilon}^t \\
&+ \chi \left[\frac{2(b''D_\lambda\psi + b'''\psi - \omega'_{0k})}{(b(y) - \lambda + i\epsilon)^3} + \frac{2(b''\partial_\lambda D_\lambda\psi + b'''\partial_\lambda\psi)}{(b(y) - \lambda + i\epsilon)^2} + \frac{b'''\partial_\lambda^2\psi}{b(y) - \lambda + i\epsilon} \right] \\
&+ 2\chi' \left[-\frac{b''D_\lambda\psi + b'''\psi - \omega'_{0k}}{b'} + \frac{b''(b''\psi - \omega_{0k})}{(b')^2} \right] \frac{1}{(b(y) - \lambda + i\epsilon)^2}.
\end{aligned}$$

(6) $\mathcal{R}_{1,2,\epsilon}^t(y, \lambda)$ appearing in (5.60) and (5.61) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{1,2,\epsilon}^t(y, \lambda) &= c_1^{(1,2)}(y) \frac{\log(b(y) - \lambda + i\epsilon)}{b(y) - \lambda + i\epsilon} + \frac{c_2^{(1,2)}(y) + c_1^{(1,2)}(y)\partial_y D_\lambda^2\psi + c_2^{(1,2)}(y)\partial_y D_\lambda\psi + c_3^{(1,2)}(y)\partial_y\psi}{b(y) - \lambda + i\epsilon} \\
&+ c_3^{(1,2)}(y) \log(b(y) - \lambda + i\epsilon) + c_4^{(1,2)}(y),
\end{aligned} \tag{A.6}$$

with

$$\begin{aligned}
c_1^{(1,2)}(y, \lambda) &= -(b')^2 c_1^{(3,0)} + 2b'c_1^{(2,1)} + b''c_1^{(2,0)}, \\
c_2^{(1,2)}(y) &= -(b')^2 c_2^{(3,0)} + 2b'c_2^{(2,1)} + b''c_2^{(2,0)} \\
&+ 2k^2 b' \left[\partial_y \left(\frac{1-\chi}{(b')^2} \right) (b''\psi - \omega_{0k}) + \frac{1-\chi}{(b')^2} (b'''\psi - \omega'_{0k}) \right] + k^2 \frac{1-\chi}{(b')^2} b''(b''\psi - \omega_{0k}), \\
c_1^{(1,2)}(y) &= -(b')^2 c_1^{(3,0)} + 2b'c_1^{(2,1)}, \\
c_2^{(1,2)}(y) &= -(b')^2 c_2^{(3,0)} + 2b'c_2^{(2,1)} + b''c_1^{(2,0)}, \\
c_3^{(1,2)}(y) &= -(b')^2 c_3^{(3,0)} + 2b'c_3^{(2,1)} + b''c_2^{(2,0)} + 2k^2 b' \frac{1-\chi}{(b')^2} b'', \\
c_3^{(1,2)}(y) &= -(b')^2 c_3^{(3,0)} + 2b'c_3^{(2,1)} + b''c_3^{(2,0)} \\
&+ k^2 \left[2\partial_y \left(\frac{(1-\chi)b''}{(b')^2} \right) \partial_y\psi + \partial_y^2 \left(\frac{(1-\chi)b''}{(b')^2} \right) \psi - \frac{1-\chi}{(b')^2} \omega'_{0k} \right] \\
&+ k^2 \left[-2\partial_y \left(\frac{1-\chi}{(b')^2} \right) \omega'_{0k} + \left(k^2 \frac{1-\chi}{(b')^2} - \partial_y^2 \left(\frac{1-\chi}{(b')^2} \right) \right) \omega_{0k} \right], \\
c_4^{(1,2)}(y) &= -2(b'')^2 r_{3,0,\epsilon}^t + 4b''\partial_y r_{2,1,\epsilon}^t + 2b'''r_{2,1,\epsilon}^t + 2b'''\partial_y r_{2,0,\epsilon}^t + b^{(4)}r_{2,0,\epsilon}^t \\
&+ \chi \left[\frac{b''D_\lambda^2\psi + 2b'''D_\lambda\psi + b^{(4)}\psi - \omega'_{0k}}{(b(y) - \lambda + i\epsilon)^2} + \frac{2b'''\partial_\lambda D_\lambda\psi + b^{(4)}\partial_\lambda\psi}{b(y) - \lambda + i\epsilon} \right] \\
&+ 2\chi' \left[\frac{b''\partial_y D_\lambda\psi + b'''\partial_\lambda\psi}{b(y) - \lambda + i\epsilon} + \frac{b''(b''\psi - \omega_{0k})}{b'(b(y) - \lambda + i\epsilon)^2} \right] \\
&+ \chi'' \frac{b''\partial_\lambda\psi}{b(y) - \lambda + i\epsilon}.
\end{aligned}$$

A.2. Degenerate case. In the degenerate case, we consider

$$\left(-\partial_y^2 + k^2 + \frac{b''(y)}{b(y) - \lambda + i\epsilon} \right) \left(\partial_\lambda^a D_\lambda^b F_{k,\epsilon}^t - \mathcal{S}_{a,b,\epsilon}^t \right) (y, \lambda) = \mathcal{R}_{a,b,\epsilon}^t,$$

where $F_{k,\epsilon}^t$ is defined in (7.1) and satisfies (7.9). We organize the terms in $\mathcal{R}_{a,b,\epsilon}$ according to their scaling dimensions. Specifically, terms whose acceptability depends on a cancellation are kept together and displayed first. The terms of non-positive scaling dimension are listed in increasing order of dimension. In this convention, k^2 has scaling dimension -2 . Terms containing r are listed next. Finally, we display the terms whose apparently stronger singularities are controlled by the accompanying cutoff factors.

Throughout this subsection, we use the following shorthand notations:

$$f = F_{k,\epsilon}^t(y, \lambda), \quad b = b(y), \quad \chi = \chi_{j,\delta}^*(y), \quad c = \delta^2.$$

(1) $\mathcal{R}_{1,0,\epsilon}'(y, \lambda)$ appearing in (7.15) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{1,0,\epsilon}' &= \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{4(b'')^2}{(b')^3} \partial_y f - \frac{6(b'')^3}{(b')^4} f \right] + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} \partial_y f + \frac{3(b'')^2}{(b')^2} f \right] && \text{order } -2 \\
&+ \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^2} \partial_y f + \frac{4(b'')^2}{(b')^3} f \right] + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} f \right] \\
&+ \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''}{(b')^2} f \right] \\
&+ \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} \partial_y f + \frac{6b''b'''}{(b')^3} f \right] + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{b'} f \right] && \text{order } -1 \\
&+ \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} f \right] \\
&+ \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} f \right] && \text{order } 0 \\
&+ \partial_\lambda r + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{b''}{(b')^2} r \right] && r\text{-terms} \\
&+ (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-b'' f \right] \right] && 1-\chi \text{ terms}
\end{aligned} \tag{A.7}$$

(2) $\mathcal{R}_{2,0,\epsilon}'(y, \lambda)$ appearing in (7.23) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{2,0,\epsilon}' &= \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[\frac{4(b'')^3}{(b')^4} f \right] && \text{order } -4 \\
&+ \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{8(b'')^2}{(b')^4} \partial_y(D_\lambda f) - \frac{20(b'')^3}{(b')^5} D_\lambda f - \frac{28(b'')^3}{(b')^5} \partial_y f + \frac{60(b'')^4}{(b')^6} f + \frac{4k^2(b'')^2}{(b')^4} f \right] \\
&+ \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{4b''}{(b')^2} \partial_y(D_\lambda f) + \frac{10(b'')^2}{(b')^3} D_\lambda f + \frac{2(b'')^2}{(b')^3} \partial_y f - \frac{15(b'')^3}{(b')^4} f \right] \\
&+ \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b''}{(b')^3} \partial_y(D_\lambda f) + \frac{12(b'')^2}{(b')^4} D_\lambda f + \frac{6(b'')^2}{(b')^4} \partial_y f - \frac{24(b'')^3}{(b')^5} f \right] \\
&+ \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{4b''}{(b')^2} D_\lambda f + \frac{2b''}{(b')^2} \partial_y f + \frac{2(b'')^2}{(b')^3} f \right] \\
&+ \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^3} D_\lambda f + \frac{3(b'')^2}{(b')^4} f \right] + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{b''}{(b')^2} f \right] \\
&+ \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{2b''b'''}{(b')^3} f \right] && \text{order } -3 \\
&+ \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b'''}{(b')^3} \partial_y(D_\lambda f) + \frac{16b''b'''}{(b')^4} D_\lambda f + \frac{26b''b'''}{(b')^4} \partial_y f - \frac{84b''''(b'')^2}{(b')^5} f \right] \\
&+ \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{4b'''}{(b')^2} D_\lambda f - \frac{2b'''}{(b')^2} \partial_y f + \frac{16b''b'''}{(b')^3} f \right] \\
&+ \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b'''}{(b')^3} D_\lambda f - \frac{4b'''}{(b')^3} \partial_y f + \frac{24b''b'''}{(b')^4} f \right] \\
&+ \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{(b')^2} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^3} f \right] \\
&+ \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^3} D_\lambda f - \frac{4b''''}{(b')^3} \partial_y f + \frac{12(b''')^2}{(b')^4} f + \frac{18b''b''''}{(b')^4} f \right] && \text{order } -2
\end{aligned}$$

$$\begin{aligned}
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b''''}{(b')^2} f \right] + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{4b''''}{(b')^3} f \right] \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b^{(5)}}{(b')^3} f \right] \quad \text{order } -1 \\
& + \partial_\lambda^2 r + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[\frac{2b''}{(b')^3} D_\lambda r - \frac{7(b'')^2}{(b')^4} r + \frac{2b'''}{(b')^3} r \right] \quad r\text{-terms} \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{b''}{(b')^2} r \right] \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^3} [-2b'' f] + \frac{1}{(b-\lambda+i\epsilon)^2} [-2b'' \partial_\lambda f] \right] \quad 1-\chi \text{ terms} \\
& + \chi' \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-\frac{2b''}{b'} f \right] \right] \quad \chi' \text{ terms}
\end{aligned} \tag{A.8}$$

(3) $\mathcal{R}_{1,1,\epsilon}^t(y, \lambda)$ appearing in (7.27) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{1,1,\epsilon}^t = & -2b'' \partial_y \left(\partial_\lambda^2 f - \chi \frac{b'' f}{(b')^2 (b-\lambda+i\epsilon)} \right) \quad \text{cancellations} \\
& + \frac{\chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right)}{(b-\lambda+i\epsilon)} \left[\frac{2(b'')^3}{(b')^3} f \right] \quad \text{order } -3 \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[\frac{2(b'')^2}{(b')^3} \partial_y (D_\lambda f) - \frac{4(b'')^3}{(b')^4} D_\lambda f - \frac{14(b'')^3}{(b')^4} \partial_y f + \frac{24(b'')^4}{(b')^5} f + \frac{2k^2(b'')^2}{(b')^3} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} \partial_y (D_\lambda f) + \frac{3(b'')^2}{(b')^2} D_\lambda f + \frac{2(b'')^2}{(b')^2} \partial_y f - \frac{6(b'')^3}{(b')^3} f \right] \\
& + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b''}{(b')^2} \partial_y (D_\lambda f) + \frac{4(b'')^2}{(b')^3} D_\lambda f + \frac{4(b'')^2}{(b')^3} \partial_y f - \frac{12(b'')^3}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} D_\lambda f + \frac{2(b'')^2}{(b')^2} f \right] \\
& + \chi'' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{b''}{(b')^2} D_\lambda f + \frac{2(b'')^2}{(b')^3} f \right] \\
& + \frac{\chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right)}{(b-\lambda+i\epsilon)} \left[-\frac{b'' b'''}{(b')^2} f \right] \quad \text{order } -2 \\
& + \chi \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b'''}{(b')^2} \partial_y (D_\lambda f) + \frac{5b'' b'''}{(b')^3} D_\lambda f + \frac{13b'' b'''}{(b')^3} \partial_y f - \frac{36b'' b'''}{(b')^4} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b'''}{b'} D_\lambda f - \frac{b'''}{b'} \partial_y f + \frac{9b'' b'''}{(b')^2} f \right] + \chi [-b''' \partial_\lambda^2 f] \\
& + \frac{(1-\chi)}{(b-\lambda+i\epsilon)} [-b''' \partial_\lambda f] + (1-\chi) [-b''' \partial_\lambda^2 f] \\
& + \chi' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{2b'''}{(b')^2} D_\lambda f - \frac{2b'''}{(b')^2} \partial_y f + \frac{12b'' b'''}{(b')^3} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{b'} f \right] \\
& + \chi'' \log \left(\frac{b-\lambda+i\epsilon}{c} \right) \left[-\frac{b'''}{(b')^2} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} D_\lambda f - \frac{2b''''}{(b')^2} \partial_y f + \frac{6(b''')^2}{(b')^3} f + \frac{8b''b''''}{(b')^3} f \right] & \text{order } -1 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{b'} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b^{(5)}}{(b')^2} f \right] & \text{order } 0 \\
& + D_\lambda \partial_\lambda r + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{b''}{(b')^2} D_\lambda r - \frac{4(b'')^2}{(b')^3} r + \frac{b'''}{(b')^2} r \right] & r\text{-terms} \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-b'' D_\lambda f - b''' f \right] \right] & 1-\chi \text{ terms}
\end{aligned} \tag{A.9}$$

(4) For $\mathcal{R}_{3,0,\epsilon}^\iota(y, \lambda)$ appearing in (7.39) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{3,0,\epsilon}^\iota = & \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[\frac{12(b'')^3}{(b')^5} D_\lambda f \right] & \text{order } -6 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{12(b'')^2}{(b')^5} \partial_y (D_\lambda^2 f) - \frac{48(b'')^3}{(b')^6} D_\lambda^2 f - \frac{132(b'')^3}{(b')^6} \partial_y (D_\lambda f) + \frac{360(b'')^4}{(b')^7} D_\lambda f \right. \\
& + \frac{12k^2(b'')^2}{(b')^5} D_\lambda f + \frac{6(b'')^3}{(b')^4} \partial_\lambda^2 f + \frac{48(b'')^3}{(b')^5} \partial_y \partial_\lambda f + \frac{180(b'')^4}{(b')^7} \partial_y f + \frac{8k^2(b'')^2}{(b')^5} \partial_y f \\
& \left. - \frac{630(b'')^5}{(b')^8} f - \frac{20k^2(b'')^3}{(b')^6} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{6b''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{21(b'')^2}{(b')^4} D_\lambda^2 f + \frac{12(b'')^2}{(b')^4} \partial_y (D_\lambda f) - \frac{78(b'')^3}{(b')^5} D_\lambda f \right. \\
& \left. - \frac{12(b'')^3}{(b')^5} \partial_y f - \frac{2b''k^2}{(b')^3} \partial_y f + \frac{105(b'')^4}{(b')^6} f + \frac{7k^2(b'')^2}{(b')^4} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b''}{(b')^4} \partial_y (D_\lambda^2 f) + \frac{24(b'')^2}{(b')^5} D_\lambda^2 f + \frac{24(b'')^2}{(b')^5} \partial_y (D_\lambda f) \right. \\
& \left. - \frac{120(b'')^3}{(b')^6} D_\lambda f - \frac{30(b'')^3}{(b')^6} \partial_y f - \frac{2b''k^2}{(b')^4} \partial_y f + \frac{180(b'')^4}{(b')^7} f + \frac{8k^2(b'')^2}{(b')^5} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{6b''}{(b')^3} D_\lambda^2 f + \frac{6b''}{(b')^3} \partial_y (D_\lambda f) + \frac{6(b'')^2}{(b')^4} D_\lambda f - \frac{6(b'')^3}{(b')^5} f - \frac{4b''k^2}{(b')^3} f \right] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{3b''}{(b')^4} D_\lambda^2 f + \frac{12(b'')^2}{(b')^5} D_\lambda f - \frac{15(b'')^3}{(b')^6} f - \frac{b''k^2}{(b')^4} f \right] \\
& + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{3b''}{(b')^3} D_\lambda f - \frac{b''}{(b')^3} \partial_y f - \frac{3(b'')^2}{(b')^4} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{6b''b'''}{(b')^4} D_\lambda f + \frac{24b'''(b'')^2}{(b')^5} f \right] & \text{order } -5 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b''''}{(b')^4} \partial_y (D_\lambda^2 f) + \frac{30b''b'''}{(b')^5} D_\lambda^2 f + \frac{102b''b'''}{(b')^5} \partial_y (D_\lambda f) \right. \\
& - \frac{420b'''(b'')^2}{(b')^6} D_\lambda f - \frac{18b''b'''}{(b')^4} \partial_y \partial_\lambda f + \frac{24b'''(b'')^2}{(b')^5} \partial_\lambda f - \frac{250b'''(b'')^2}{(b')^6} \partial_y f \\
& - \frac{2b'''k^2}{(b')^4} \partial_y f + \frac{1110b'''(b'')^3}{(b')^7} f + \frac{12b''b'''k^2}{(b')^5} f \left. \right] + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{6b''''}{(b')^3} D_\lambda^2 f \right. \\
& \left. - \frac{6b''''}{(b')^3} \partial_y (D_\lambda f) + \frac{66b''b'''}{(b')^4} D_\lambda f + \frac{14b''b'''}{(b')^4} \partial_y f - \frac{150b'''(b'')^2}{(b')^5} f - \frac{4b'''k^2}{(b')^3} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b'''}{(b')^4} D_\lambda^2 f - \frac{12b'''}{(b')^4} \partial_y(D_\lambda f) + \frac{96b''b'''}{(b')^5} D_\lambda f + \frac{32b''b'''}{(b')^5} \partial_y f \right. \\
& \quad \left. - \frac{250b''(b')^2}{(b')^6} f - \frac{2b''k^2}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{6b'''}{(b')^3} D_\lambda f + \frac{2b'''}{(b')^3} \partial_y f + \frac{8b''b'''}{(b')^4} f \right] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b'''}{(b')^4} D_\lambda f + \frac{16b''b'''}{(b')^5} f \right] + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{2b'''}{(b')^3} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{6(b'')^2}{(b')^4} f - \frac{3b''b'''}{(b')^4} f \right] \tag{order -4} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{3b''''}{(b')^4} D_\lambda^2 f - \frac{12b''''}{(b')^4} \partial_y(D_\lambda f) + \frac{48(b''')^2}{(b')^5} D_\lambda f + \frac{72b''b''''}{(b')^5} D_\lambda f \right. \\
& \quad - \frac{6(b'')^2}{(b')^4} \partial_\lambda f - \frac{3b''b''''}{(b')^4} \partial_\lambda f + \frac{32(b'')^2}{(b')^5} \partial_y f + \frac{56b''b''''}{(b')^5} \partial_y f - \frac{330b''(b'')^2}{(b')^6} f \\
& \quad \left. - \frac{265b''''(b'')^2}{(b')^6} f - \frac{b''''k^2}{(b')^4} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{9b''''}{(b')^3} D_\lambda f - \frac{3b''''}{(b')^3} \partial_y f + \frac{20(b''')^2}{(b')^4} f + \frac{35b''b''''}{(b')^4} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{12b''''}{(b')^4} D_\lambda f - \frac{6b''''}{(b')^4} \partial_y f + \frac{32(b'')^2}{(b')^5} f + \frac{56b''b''''}{(b')^5} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{(b')^3} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{3b''''}{(b')^4} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b^{(5)}}{(b')^4} D_\lambda f - \frac{6b^{(5)}}{(b')^4} \partial_y f + \frac{40b''b^{(5)}}{(b')^5} f + \frac{60b''b''''}{(b')^5} f \right] \tag{order -3} \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{4b^{(5)}}{(b')^3} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{6b^{(5)}}{(b')^4} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{3b^{(6)}}{(b')^4} f \right] \tag{order -2} \\
& + \partial_\lambda^3 r \tag{r-terms} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{3b''}{(b')^4} D_\lambda^2 r - \frac{24(b'')^2}{(b')^5} D_\lambda r + \frac{6b'''}{(b')^4} D_\lambda r + \frac{15(b'')^3}{(b')^6} r \right. \\
& \quad \left. + \frac{3b''''}{(b')^4} r - \frac{16b''b'''}{(b')^5} r + \frac{b''k^2}{(b')^4} r \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b''}{(b')^3} D_\lambda r + \frac{b''}{(b')^3} \partial_y r + \frac{3(b'')^2}{(b')^4} r \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[\frac{2b''}{(b')^3} r \right] \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^4} [-6b''f] + \frac{1}{(b-\lambda+i\epsilon)^3} [-6b''\partial_\lambda f] \right. \\
& \quad \left. + \frac{1}{(b-\lambda+i\epsilon)^2} [-3b''\partial_\lambda^2 f] \right] \tag{1 - \chi terms} \\
& + \chi' \left[\frac{1}{(b-\lambda+i\epsilon)^3} \left[-\frac{4b''}{b'} f \right] + \frac{1}{(b-\lambda+i\epsilon)^2} \left[-\frac{6b''}{(b')^2} D_\lambda f + \frac{4b''}{(b')^2} \partial_y f - \frac{2b'''}{(b')^2} f \right] \right] \tag{\chi' terms}
\end{aligned}$$

$$+ \chi'' \left[\frac{1}{(b - \lambda + i\epsilon)^2} \left[\frac{b''}{(b')^2} f \right] \right]. \quad \chi'' \text{ terms} \quad (\text{A.10})$$

(5) For $\mathcal{R}_{2,1,\epsilon}^t(y, \lambda)$ appearing in (7.46) has the explicit representation

$$\begin{aligned} \mathcal{R}_{2,1,\epsilon}^t = & -2b'' \partial_y (\partial_\lambda^3 f - \chi S_{3,0}) - b''' (\partial_\lambda^3 f - \chi S_{3,0}) - \frac{b'''}{(b - \lambda + i\epsilon)} (\partial_\lambda^2 f - \chi S_{2,0}) \quad \text{cancellations} \\ & - \frac{8\chi(b'')^3}{(b')^3} \partial_\lambda^2 f \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \\ & + \frac{\chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right)}{(b - \lambda + i\epsilon)} \left[\frac{8(b'')^3}{(b')^4} D_\lambda f - \frac{22(b'')^4}{(b')^5} f \right] \quad \text{order } -5 \\ & + \chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[\frac{10(b'')^2}{(b')^4} \partial_y (D_\lambda^2 f) - \frac{28(b'')^3}{(b')^5} D_\lambda^2 f - \frac{94(b'')^3}{(b')^5} \partial_y (D_\lambda f) + \frac{250(b'')^4}{(b')^6} D_\lambda f \right. \\ & \quad + \frac{8k^2(b'')^2}{(b')^4} D_\lambda f + \frac{28(b'')^4}{(b')^5} \partial_\lambda f + \frac{200(b'')^4}{(b')^6} \partial_y f + \frac{2k^2(b'')^2}{(b')^4} \partial_y f - \frac{540(b'')^5}{(b')^7} f \\ & \quad \left. - \frac{30k^2(b'')^3}{(b')^5} f \right] \\ & + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} \partial_y (D_\lambda^2 f) + \frac{16(b'')^2}{(b')^3} D_\lambda^2 f + \frac{10(b'')^2}{(b')^3} \partial_y (D_\lambda f) \right. \\ & \quad \left. - \frac{63(b'')^3}{(b')^4} D_\lambda f - \frac{2(b'')^3}{(b')^3} \partial_\lambda f - \frac{12(b'')^3}{(b')^4} \partial_y f + \frac{90(b'')^4}{(b')^5} f + \frac{2k^2(b'')^2}{(b')^3} f \right] \\ & + \chi' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{18(b'')^2}{(b')^4} D_\lambda^2 f + \frac{18(b'')^2}{(b')^4} \partial_y (D_\lambda f) - \frac{96(b'')^3}{(b')^5} D_\lambda f \right. \\ & \quad \left. - \frac{24(b'')^3}{(b')^5} \partial_y f + \frac{150(b'')^4}{(b')^6} f + \frac{2k^2(b'')^2}{(b')^4} f \right] \\ & + \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{4b''}{(b')^2} D_\lambda^2 f + \frac{2b''}{(b')^2} \partial_y (D_\lambda f) + \frac{8(b'')^2}{(b')^3} D_\lambda f - \frac{2(b'')^2}{(b')^3} \partial_y f - \frac{6(b'')^3}{(b')^4} f \right] \\ & + \chi'' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{2b''}{(b')^3} D_\lambda^2 f + \frac{9(b'')^2}{(b')^4} D_\lambda f - \frac{12(b'')^3}{(b')^5} f \right] \\ & + \frac{\chi''}{(b - \lambda + i\epsilon)} \left[\frac{b''}{(b')^2} D_\lambda f - \frac{2(b'')^2}{(b')^3} f \right] \\ & + \frac{\chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right)}{(b - \lambda + i\epsilon)} \left[-\frac{2b''b'''}{(b')^3} D_\lambda f + \frac{26b''b'''}{(b')^4} f \right] \quad \text{order } -4 \\ & + \chi \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b'''}{(b')^3} \partial_y (D_\lambda^2 f) + \frac{23b''b'''}{(b')^4} D_\lambda^2 f + \frac{64b''b'''}{(b')^4} \partial_y (D_\lambda f) \right. \\ & \quad \left. - \frac{285b''b'''}{(b')^5} D_\lambda f - \frac{22b''b'''}{(b')^4} \partial_\lambda f - \frac{251b''b'''}{(b')^5} \partial_y f + \frac{925b''b'''}{(b')^6} f + \frac{15b''b''b''k^2}{(b')^4} f \right] \\ & + \frac{\chi}{(b - \lambda + i\epsilon)} \left[-\frac{4b'''}{(b')^2} D_\lambda^2 f - \frac{6b'''}{(b')^2} \partial_y (D_\lambda f) + \frac{52b''b'''}{(b')^3} D_\lambda f \right. \\ & \quad \left. + \frac{2b''b'''}{(b')^2} \partial_\lambda f + \frac{12b''b'''}{(b')^3} \partial_y f - \frac{125b''b'''}{(b')^4} f \right] \\ & + \chi' \log \left(\frac{b - \lambda + i\epsilon}{c} \right) \left[-\frac{4b'''}{(b')^3} D_\lambda^2 f - \frac{8b'''}{(b')^3} \partial_y (D_\lambda f) + \frac{72b''b'''}{(b')^4} D_\lambda f + \frac{24b''b'''}{(b')^4} \partial_y f - \frac{200b''b''b''k^2}{(b')^5} f \right] \\ & + \frac{\chi'}{(b - \lambda + i\epsilon)} \left[-\frac{6b'''}{(b')^2} D_\lambda f + \frac{2b'''}{(b')^2} \partial_y f + \frac{8b''b'''}{(b')^3} f \right] \end{aligned}$$

$$\begin{aligned}
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b'''}{(b')^3} D_\lambda f + \frac{12b''b'''}{(b')^4} f \right] + \frac{\chi''}{(b-\lambda+i\epsilon)} \left[\frac{b'''}{(b')^2} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{2(b''')^2}{(b')^3} f - \frac{2b''b'''}{(b')^3} f \right] \quad \text{order } -3 \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^3} D_\lambda^2 f - \frac{8b''''}{(b')^3} \partial_y(D_\lambda f) + \frac{32(b''')^2}{(b')^4} D_\lambda f + \frac{48b''b''''}{(b')^4} D_\lambda f \right. \\
& \quad \left. + \frac{4b''b''''}{(b')^3} \partial_\lambda f + \frac{28(b''')^2}{(b')^4} \partial_y f + \frac{48b''b''''}{(b')^4} \partial_y f - \frac{264b''(b''')^2}{(b')^5} f - \frac{212b''''(b'')^2}{(b')^5} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{7b''''}{(b')^2} D_\lambda f - \frac{2b''''}{(b')^2} \partial_y f + \frac{16(b''')^2}{(b')^3} f + \frac{28b''b''''}{(b')^3} f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{8b''''}{(b')^3} D_\lambda f - \frac{4b''''}{(b')^3} \partial_y f + \frac{24(b''')^2}{(b')^4} f + \frac{42b''b''''}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{(b')^2} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''''}{(b')^3} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b^{(5)}}{(b')^3} D_\lambda f - \frac{4b^{(5)}}{(b')^3} \partial_y f + \frac{30b''b^{(5)}}{(b')^4} f + \frac{45b''''b^{(5)}}{(b')^4} f \right] \quad \text{order } -2 \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{3b^{(5)}}{(b')^2} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b^{(5)}}{(b')^3} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b^{(6)}}{(b')^3} f \right] \quad \text{order } -1 \\
& + D_\lambda \partial_\lambda^2 r \quad \text{r-terms} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{2b''}{(b')^3} D_\lambda^2 r - \frac{17(b'')^2}{(b')^4} D_\lambda r + \frac{4b''''}{(b')^3} D_\lambda r + \frac{34(b'')^3}{(b')^5} r + \frac{2b''''}{(b')^3} r - \frac{24b''b''''}{(b')^4} r \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{b''}{(b')^2} D_\lambda r + \frac{2(b'')^2}{(b')^3} r - \frac{b''''}{(b')^2} r \right] \\
& + (1-\chi) \left[\frac{1}{(b-\lambda+i\epsilon)^3} \left[-2b'' D_\lambda f - 2b''' f \right] + \frac{1}{(b-\lambda+i\epsilon)^2} \left[-2b'' \partial_\lambda D_\lambda f - 2b''' \partial_\lambda f \right] \right] \quad 1-\chi \text{ terms} \\
& + \chi' \left[\frac{1}{(b-\lambda+i\epsilon)^2} \left[-\frac{2b''}{b'} D_\lambda f + \frac{2(b'')^2}{(b')^2} f - \frac{2b''''}{b'} f \right] \right] \quad \chi' \text{ terms}
\end{aligned} \tag{A.11}$$

(6) For $\mathcal{R}_{1,2,\epsilon}^\ell(y, \lambda)$ appearing in (7.53) has the explicit representation

$$\begin{aligned}
\mathcal{R}_{1,2,\epsilon}^\ell = & -4b'' \partial_y (\partial_\lambda^2 D_\lambda f - \chi S_{2,1}) + 2(b'')^2 (\partial_\lambda^3 f - \chi S_{3,0}) - 2b''' \partial_y (\partial_\lambda^2 f - \chi S_{2,0}) \quad \text{cancellations} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[\frac{12(b'')^2}{(b')^3} \partial_y (D_\lambda^2 f) - \frac{36(b'')^3}{(b')^4} D_\lambda^2 f - \frac{60(b'')^3}{(b')^4} \partial_y (D_\lambda f) + \frac{216(b'')^4}{(b')^5} D_\lambda f \right. \quad \text{order } -4 \\
& \quad - \frac{4(b'')^2}{(b')^2} \partial_y \partial_\lambda (D_\lambda f) + \frac{2(b'')^3}{(b')^2} \partial_\lambda^2 f + \frac{8(b'')^3}{(b')^3} \partial_y \partial_\lambda f + \frac{96(b'')^4}{(b')^5} \partial_y f - \frac{390(b'')^5}{(b')^6} f \\
& \quad \left. - \frac{2k^2(b'')^3}{(b')^4} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} \partial_y (D_\lambda^2 f) + \frac{11(b'')^2}{(b')^2} D_\lambda^2 f + \frac{4(b'')^2}{(b')^2} \partial_y (D_\lambda f) \right. \\
& \quad \left. - \frac{42(b'')^3}{(b')^3} D_\lambda f - \frac{6(b'')^3}{(b')^3} \partial_y f + \frac{60(b'')^4}{(b')^4} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b''}{(b')^2} \partial_y(D_\lambda^2 f) + \frac{12(b'')^2}{(b')^3} D_\lambda^2 f + \frac{8(b'')^2}{(b')^3} \partial_y(D_\lambda f) \right. \\
& \quad \left. - \frac{60(b'')^3}{(b')^4} D_\lambda f - \frac{12(b'')^3}{(b')^4} \partial_y f + \frac{96(b'')^4}{(b')^5} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''}{b'} D_\lambda^2 f + \frac{4(b'')^2}{(b')^2} D_\lambda f - \frac{4(b'')^3}{(b')^3} f \right] \\
& + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''}{(b')^2} D_\lambda^2 f + \frac{4(b'')^2}{(b')^3} D_\lambda f - \frac{6(b'')^3}{(b')^4} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{2b''b'''}{(b')^2} D_\lambda f + \frac{4b'''(b'')^2}{(b')^3} f \right] \tag{order -3} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} \partial_y(D_\lambda^2 f) + \frac{14b''b'''}{(b')^3} D_\lambda^2 f + \frac{44b''b'''}{(b')^3} \partial_y(D_\lambda f) \right. \\
& \quad - \frac{216b'''(b'')^2}{(b')^4} D_\lambda f - \frac{2b''b'''}{(b')^2} \partial_\lambda D_\lambda f - \frac{6b''b'''}{(b')^2} \partial_y \partial_\lambda f + \frac{4b'''(b'')^2}{(b')^3} \partial_\lambda f \\
& \quad \left. - \frac{126b'''(b'')^2}{(b')^4} \partial_y f + \frac{632b'''(b'')^3}{(b')^5} f \right] \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b'''}{b'} D_\lambda^2 f - \frac{4b'''}{b'} \partial_y(D_\lambda f) + \frac{38b''b'''}{(b')^2} D_\lambda f - 2b''' \partial_\lambda D_\lambda f + \frac{6b''b'''}{(b')^2} \partial_y f \right. \\
& \quad \left. - \frac{88b'''(b'')^2}{(b')^3} f \right] + \chi \left[-2b''' \partial_\lambda^2 D_\lambda f \right] \\
& + \frac{(1-\chi)}{(b-\lambda+i\epsilon)} \left[-2b''' \partial_\lambda D_\lambda f \right] + (1-\chi) \left[-2b''' \partial_\lambda^2 D_\lambda f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} D_\lambda^2 f \right. \\
& \quad \left. - \frac{4b'''}{(b')^2} \partial_y(D_\lambda f) + \frac{44b''b'''}{(b')^3} D_\lambda f + \frac{12b''b'''}{(b')^3} \partial_y f - \frac{126b'''(b'')^2}{(b')^4} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{4b'''}{b'} D_\lambda f + \frac{6b''b'''}{(b')^2} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b'''}{(b')^2} D_\lambda f + \frac{6b''b'''}{(b')^3} f \right] \\
& + \frac{\chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right)}{(b-\lambda+i\epsilon)} \left[-\frac{2(b''')^2}{(b')^2} f - \frac{b''b'''}{(b')^2} f \right] \tag{order -2} \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} D_\lambda^2 f - \frac{4b''''}{(b')^2} \partial_y(D_\lambda f) + \frac{16(b''')^2}{(b')^3} D_\lambda f + \frac{32b''b''''}{(b')^3} D_\lambda f \right. \\
& \quad - \frac{2(b''')^2}{(b')^2} \partial_\lambda f - \frac{b''b''''}{(b')^2} \partial_\lambda f + \frac{16(b''')^2}{(b')^3} \partial_y f + \frac{24b''b''''}{(b')^3} \partial_y f - \frac{162b''(b''')^2}{(b')^4} f \\
& \quad - \frac{138b''''(b'')^2}{(b')^4} f \left. \right] + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{4b''''}{b'} D_\lambda f - b'''' \partial_\lambda f - \frac{2b''''}{b'} \partial_y f + \frac{14(b''')^2}{(b')^2} f \right. \\
& \quad \left. + \frac{19b''b''''}{(b')^2} f \right] + \chi \left[-b'''' \partial_\lambda^2 f \right] + \frac{(1-\chi)}{(b-\lambda+i\epsilon)} \left[-b'''' \partial_\lambda f \right] + (1-\chi) \left[-b'''' \partial_\lambda^2 f \right] \\
& + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{4b''''}{(b')^2} D_\lambda f - \frac{2b''''}{(b')^2} \partial_y f + \frac{16(b''')^2}{(b')^3} f + \frac{24b''b''''}{(b')^3} f \right] \\
& + \frac{\chi'}{(b-\lambda+i\epsilon)} \left[-\frac{2b''''}{b'} f \right] + \chi'' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{b''''}{(b')^2} f \right] \\
& + \chi \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b^{(5)}}{(b')^2} D_\lambda f - \frac{2b^{(5)}}{(b')^2} \partial_y f + \frac{18b''b^{(5)}}{(b')^3} f + \frac{24b''''b''''}{(b')^3} f \right] \tag{order -1} \\
& + \frac{\chi}{(b-\lambda+i\epsilon)} \left[-\frac{2b^{(5)}}{b'} f \right] + \chi' \log\left(\frac{b-\lambda+i\epsilon}{c}\right) \left[-\frac{2b^{(5)}}{(b')^2} f \right]
\end{aligned}$$

$$\begin{aligned}
& + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[-\frac{b^{(6)}}{(b')^2} f\right] && \text{order 0} \\
& + D_\lambda^2 \partial_\lambda r + \chi \log\left(\frac{b - \lambda + i\epsilon}{c}\right) \left[\frac{b''}{(b')^2} D_\lambda^2 r - \frac{4(b'')^2}{(b')^3} D_\lambda r + \frac{2b'''}{(b')^2} D_\lambda r + \frac{6(b'')^3}{(b')^4} r \right. \\
& \quad \left. + \frac{b''''}{(b')^2} r - \frac{6b''b'''}{(b')^3} r\right] && r\text{-terms} \\
& + (1 - \chi) \left[\frac{1}{(b - \lambda + i\epsilon)^2} \left[-b'' D_\lambda^2 f - 2b''' D_\lambda f - b'''' f\right]\right]. && 1 - \chi \text{ terms}
\end{aligned}
\tag{A.12}$$

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